

강의계획서

2026학년도 1학기

과목명	일반생물학실험		교과목 번호	CB3202	학점	1	강의실	탐501
			교과 구분	AP, Elective	시수	2		
담당교원	이름	분반	Office	Office hour	연락처		이메일	
	김정수	1(수2,3)	창402	수3, 목6	051-606-2320	kjs1@pusan.ac.kr		
		2(목7,8)						

1. 교과목 개요

This course aims to help students understand the principles of DNA cloning, which is the basic method of life science and biotechnology, and to develop the ability to apply the technique to related research topics. It is designed to actually experience a single biological research project by connecting individual experiments, including genomic DNA isolation, DNA electrophoresis, PCR, DNA ligation, E. coli transformation, plasmid purification, and restriction enzyme digestion. In addition, students can be familiar with the safety rules by conducting experiments in a biology laboratory. Through this class, students will be able to understand the characteristics of carrying out biology research projects and use DNA cloning technology for future research activities.

2. 중점 목표 역량

Subject Core Competencies						
Scientific knowledge comprehension	Critical thinking ability	Problem-solving skill	Ability to collaborate	Complex communication skills	Scientific inquiry	Creativity
In this course, students can develop communication skills, collaboration skills, and scientific inquiry while performing experiments in teams. In addition, critical thinking and creativity can be cultivated through the process of analyzing and discussing experimental data by documenting lab notebook. Through pre-lab reports, worksheet activity and group presentations, students will be able to enhance scientific knowledge comprehension, collaboration skills and communication skills.						

3. 교재 및 참고자료

- Textbook: A protocol and lab notebook for the general biology experiment will be provided.
- Reference: Biology, a global approach, 12th edition, Campbell *et al.*

4. 수업 방법

- Collaborative learning method: The class will be centered on collaborative (or cooperative) learning.
- Lecture method: Lectures are provided at the level necessary for conducting the experiment.

- ※ Bring your lab coat, lab notebook, and a fine waterproof marker for use in the lab.
- ※ For assignments and notifications, join the Google Classroom using the class code from your teacher.

5. 평가 기준 및 방법

A. 평가 기준

Lab Work	Homework (Pre-lab report)	Post-lab report	Group Presentation	Attendance and Attitude	Total
30	20	20	20	10	100

B. 평가 방법

- Lab Work
 - This grade component is based on your performance, cooperativity, and clean-up after work.
 - Good performance and experimental results will gain a point.
- Homework (Pre-lab report)
 - In the weekly plan, each homework is noted with page numbers (*H/W).
- Post-lab report
 - In the weekly plan, each post-lab report is noted with page numbers (*H/W).
 - Write the report with appropriate units (ml, µl, °C and etc.)
- Group Presentation
 - The purpose of the group presentation is to ensure that you understand the whole project of 'general biology lab' and you have a chance to study related lab methods further with other group members.
 - During the experiment, take pictures of the tool or process. This will help you to make the PPT or poster.
- Attendance and Attitude
 - Attendance in the lab is mandatory.
 - An absence will cost 2 points. If you come late, it will cost 0.5 points. Be on time.

6. 학점 부여 기준

☞ 학업 성취도가 우수한 학생은 과목우수상으로 추천하고, 교과 세부 특기 사항에 핵심 역량을 포함하여 교과 활동과 관련된 세부능력, 수행평가 및 학습활동 참여도, 태도 등을 기록함. 단, 학업 태도나 출결이 좋지 않은 학생은 제외함

- 학점 취득을 위해서는 총 수업 시간의 3/4 이상 출석이 필수임.
- 결사자 인정점 부여를 위한 근거 점수 산출 방식 :
 - 공결이나 병결로 인해 결사한 시험의 인정점은 (학생이 응시한 동일 과목의 다른 시험에서 받은 점수의 총합의 백분 위에 해당하는 결사 시험의 동일 백분위 점수) 곱하기(공결: 100%, 병결: 80%)로 한다. (※ 인정점 반영 비율은 '학업성적관리규정' 참조)
 - 본 과목에서는 자필고사(중간/기말고사)가 없으므로, (가)의 방식을 따르지 않는다. 그 대신에 공결이나 병결이 있는 경우에는 사전이나 사후에 보고서(Pre-lab, post-lab report)를 제출해야 한다. 이를 지키지 않을 경우는 해당 누락 분에 대하여 감점을 받을 수 있다.

6. 주별 강의 계획

Week (period)	Topic and Contents	Notes
Week 1 2/25-2/26	1. Introduction to the course 2. Laboratory safety 3. Guidelines for a lab notebook	Pre-class survey, Lecture for class overview
Core Competencies	♦ Complex communication skill	
Week 2 3/4-3/5	4. Laboratory equipment and terminology 5. Use of Micropipette and Centrifugation (5.1. Micropipette)	Post-lab report (p.14,15)
Core Competencies	♦ Complex communication skill, Scientific knowledge comprehension	
Week 3 3/11-3/12	6. Preparation of LB and MRS media and Sterilization (make LB and MRS liquid and solid media) 5. Use of Micropipette and Centrifugation (5.2. Centrifugation) * During autoclave, do 5.2. Centrifugation	Post-lab report (p.21)
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 4 3/18-3/19	7. Preparation of a Streak Plate for Single Colonies -research about lactic acid bacteria * Transfer the plates to the 4°C fridge after 2-3 days (when colonies are grown) 8. Picking a colony (<u>*one day before next class</u>) * Make 0.8% agarose gels (p.40) for lab #12 & lab #18	*H/W: Pre-lab report (p.29) due next class
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 5 3/25-3/26	9. Isolation of Genomic DNA from Bacteria	Post-lab reports (p.24, 25, 30) *H/W: Pre-lab report (p.37) due next class
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 6 4/1-4/2	10. Measuring DNA Concentration by Spectrophotometer 11. PCR of 16S rDNA	Post-lab report (p.33) *H/W: Pre-lab report (p.42) due next class
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 7 4/8-4/9	12. DNA electrophoresis	Post-lab report (p.43) *H/W: Pre-lab reports (p.45,50) due next class
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 8 4/15-4/16	<i>10/6-7 Chuseok Holiday 10/8 (Wed) Substitute Holiday 10/9 (Thu) Hangul Proclamation Day 10/10 (Fri) Discretionary holiday</i>	
Week 9 4/22-4/23	Midterm Exams (No midterm exam for this course)	

Week (period)	Topic and Contents	Notes
Week 10 4/29-4/30	13. PCR product purification 14. DNA Ligation	*H/W: Pre-lab report (p.52) due next class
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 11 5/6-5/7	15. <i>E. coli</i> Transformation *Transfer the plates to the 4°C fridge next day *1 day prior to the next week, inoculate bacterial colony <i>10/31(Fri)-11/2 SAC</i>	*H/W: Pre-lab reports (p.55,58) due next class
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 12 5/13-5/14	16. Purification of Bacterial Plasmid 17. Restriction enzyme digestion	Post-lab report (p.53)
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 13 5/20-5/21	18. Agarose gel electrophoresis of restricted plasmids	Post-lab report(p.61)
Core Competencies	♦ Scientific inquiry, Critical thinking, Ability to collaborate	
Week 14 5/27-5/28	19. Sanger's DNA sequencing method (theory) 20. DNA analysis using BLAST	Post-lab report(p.66) Bring your laptop
Core Competencies	♦ Scientific knowledge comprehension, Problem solving skill,	
Week 15 6/3-6/4	Preparation for group presentation (PPT or Poster)	Bring your laptop
Core Competencies	♦ Complex communication skill, Ability to collaborate, Creativity,	
Week 16 6/10-6/11	Group presentation	Bring your laptop
Week 17 6/17-6/18	Final Exams (No final exam for this course)	

※ 학사 일정에 따라 강의 계획이 변경될 수 있음